

Cheatsheet for UPnP challenge

Resources

1. **UPnP overview:** <https://upnp.org/specs/arch/UPnP-arch-DeviceArchitecture-v1.0-20080424.pdf>
2. **UPnP technical basics:** https://upnp.org/resources/documents/UPnP_UDA_tutorial_July2014.pdf

UPnP basics

Simple Service Discovery Protocol (SSDP)

- **Role:** discovery mechanism
- **Function:** allows a new device to announce its presence to the network or a control point to search for available devices (via requests over UDP/1900)
- **Output:** returns basic device info and a pointer (URL) to the root device description file
- this request will usually be multicast, but we will be using a unicast in this challenge (directly to the challenge server)

Service Control Protocol Description (SCPD)

- **Role:** The service description mechanism.
- **Function:** An XML-based file fetched during a later phase of a UPnP session to inspect a specific service
- **Output:** Details the exact list of actions, arguments, and state variables available for that service, acting similarly to an API definition before commands are sent via SOAP

1. SSDP request

A SSDP request will look like (UDP/1900):

```
M-SEARCH * HTTP/1.1
HOST: <TARGET-IP>:1900
MAN: "ssdp:discover"
MX: 2
ST: ssdp:all
```

- replace **<TARGET-IP>** with the target IP address

Example response:

```
HTTP/1.1 200 OK
LOCATION: http://<TARGET-IP>:5000/device.xml
ST: upnp:rootdevice
USN: uuid:sentinel-gateway::upnp:rootdevice
```

- take note of the **LOCATION** field
- this value URL will be used to retrieve **device.xml** in the next step (via a GET request)

You may use the provided **m-search.py** script to send the M-SEARCH request:

modify the **TARGET** variable in the provided script

```
python3 m-search.py
```

2. GET device.xml

Contains the following important information:

- **URLBase**
- **SCPDURL**
- **controlURL**
- **eventSubURL**

GET /device.xml (TCP/5000 - from previous response):

```
curl http://TARGET-IP:5000/device.xml
```

- replace **TARGET-IP** with the target IP address

Example response:

```
<URLBase>http://TARGET-IP:5000/</URLBase>

<device>
  <deviceType>
    urn:schemas-upnp-org:device:InternetGatewayDevice:1
  </deviceType>

  <service>
    <serviceType>
      urn:sentinel-org:service:DeviceConfig:1
    </serviceType>

    <serviceId>
      urn:sentinel-org:serviceId:DeviceConfig
    </serviceId>

    <SCPDURL>
      /deviceconfig.xml
    </SCPDURL>

    <controlURL>
      /upnp/control/deviceconfig
```

```
</controlURL>

<eventSubURL>
  /upnp/event/deviceconfig
</eventSubURL>
</service>

</device>
```

- **URLBase** -> **http://TARGET-IP:5000/**
- **SCPDURL** -> **/deviceconfig.xml**
- **controlURL** -> **/upnp/control/deviceconfig**
- **eventSubURL** -> **/upnp/event/deviceconfig** (NOT used)

3. GET deviceconfig.xml

Send a HTTP GET request to the **URLBase** + **SCDPURL** found earlier:

```
curl http://TARGET-IP:5000/deviceconfig.xml
```

Example response:

```
<scpd xmlns="urn:schemas-upnp-org:service-1-0">

  <actionList>
    <action>
      <name>X</name>

      <argumentList>
        <argument>
          <name>Status</name>
          <direction>out</direction>
          <relatedStateVariable>Status</relatedStateVariable>
        </argument>
      </argumentList>
    </action>
    <action>
      <name>Y</name>

      <argumentList>
        <argument>
          <name>Target</name>
          <direction>in</direction>
          <relatedStateVariable>Target</relatedStateVariable>
        </argument>

        <argument>
          <name>Result</name>
```

```

        <direction>out</direction>
        <relatedStateVariable>Result</relatedStateVariable>
    </argument>
</argumentList>

</action>
</actionList>

<serviceStateTable>

    <stateVariable sendEvents="no">
        <name>Status</name>

        <dataType>string</dataType>

    </stateVariable>

    <stateVariable sendEvents="no">
        <name>Target</name>

        <dataType>string</dataType>

        <allowedValueList>
            <allowedValue>bits</allowedValue>
            <allowedValue>bytes</allowedValue>
        </allowedValueList>

    </stateVariable>

</serviceStateTable>

</scpd>

```

Contains the following important information:

An arrow (->) will be used to indicate nested levels of tags within the XML

eg. a -> b -> c indicates that c is nested within b, and b is nested within a

- **action.name** (actionList -> action -> name)
 - used in SOAP request later on
- **argument.name** (actionList -> action -> argumentList -> name)
 - used in SOAP request later on
- **argument.direction** (actionList -> action -> argumentList -> direction)
 - indicates if its an argument to be sent, or retrieved from the server
- **argument.relatedStateVariable** (actionList -> action -> argumentList -> relatedStateVariable)
 - we can look for this value in the **stateVariable.name** down below

- `stateVariable.name` (serviceStateTable -> stateVariable -> name)
- `stateVariable.dataType` (serviceStateTable -> stateVariable -> dataType)
- `stateVariable.allowedValueList` (serviceStateTable -> stateVariable -> allowedValueList)

4. Enumerate SOAP actions

From the `deviceconfig.xml` response, we can see 2 different possible actions: **X** and **Y**

- the corresponding arguments are **Status**, **Target** + **Result** respectively
- from the nested values under `serviceStateTable`, we know:
 - **Status** -> string
 - **Target** -> string ("bits" or "bytes")

5. Send POST SOAP requests

We can now send a SOAP request to the path we retrieved from `controlURL` previously (`/upnp/control/deviceconfig`)

- this will be handled by the script

With the `soap.py` script:

```
python3 soap.py HOST --port PORT --action ACTION --arg FIELD=VALUE
# eg.
python3 soap.py TARGET-IP --port 5000 --action Y --arg Target=bytes
```

Example response:

```
Status: 200
<?xml version="1.0"?>
<s:Envelope
  xmlns:s="http://schemas.xmlsoap.org/soap/envelope/"
  s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
  <s:Body>
    <u:Y
      xmlns:u="urn:schemas-upnp-org:service:DeviceConfig:1">
      <Result>Y successfully set to bytes!</Result>
    </u:Y>
  </s:Body>
</s:Envelope>
```

Python UPnP tools

```
# M-SEARCH
```

```
python3 m-search.py
```

```
# SOAP request
```

```
python3 soap.py HOST --help
```

```
python3 soap.py HOST --port PORT --action ACTION --arg FIELD=VALUE
```

```
...
```